

Feasibility Assessment Checklist

Data Wish List Framework + 5-Dimension Evaluation

DDSI Decision Design Assistant — Bootcamp Reference

Purpose

This document provides a structured framework for evaluating whether a proposed analytical use case is feasible within given delivery timelines, data availability, and operational constraints. Use after a problem has been categorized (see Problem Taxonomy Guide) and before method selection begins.

Work through each dimension with the requesting stakeholder. A use case may still be feasible even with gaps — the goal is to identify risks early and set realistic expectations.

Dimension 1: Data Requirements — The "Data Wish List"

This is the most critical dimension. Before asking what data the stakeholder **has**, first identify what data **would be needed** to solve this problem ideally. This brainstorming step ensures the team doesn't unconsciously narrow the solution based on what's already available.

Step 1: Brainstorm Ideal Data Types (the Wish List)

Based on the problem category, proactively suggest data types that are typically needed. The agent should lead this brainstorming, not wait for the stakeholder to list data.

Problem Category	Typically Needed Data Types
Hypothesis Testing	Outcome metric data for treatment/control, pre/post measurements, sample size info, confounding variables
Regression & Correlation	Dependent variable observations, candidate independent variables, control variables, sufficient sample across range
Time Series & Forecasting	Historical metric at consistent intervals, calendar/event data, external factors (weather, economic), structural changes
Estimation & CIs	Representative sample data, population metadata, stratification variables
Distribution Fitting	Raw observation data (not pre-aggregated), sufficient sample, data collection methodology docs
Comparative Analysis	Group membership data, outcome metrics per group, demographic/contextual variables
Sampling & Survey Design	Target population definition, existing sampling frames, prior study results for power calcs
Bayesian Estimation	Historical/prior study data, expert judgment documentation, current observations (even limited)

Ask: "For this type of problem, you'd ideally want [list]. What else is relevant to your context?"

Step 2: Assess Availability Against the Wish List

- Do you currently have access to this data?
- What format — structured database, spreadsheets, reports, unstructured?
- How far back does it go? Is that enough?
- Known quality issues — missing values, inconsistencies, gaps?
- What granularity — daily, weekly, individual-level, aggregate?

Step 3: Identify Gaps and Alternatives

- Note each gap from the wish list
- Ask: "Is there a proxy or alternative source?"
- Ask: "Could this data be collected? How long?"

- Flag whether the gap is a **blocker** or a **degradation** (proceed with reduced accuracy)

Feasibility Signals — Data

- GREEN: All critical data available, accessible, sufficient history, acceptable quality
- YELLOW: Most data available but with gaps; proxies exist or quality issues manageable
- RED: Critical data missing with no viable alternatives

Dimension 2: Timeline & Delivery Constraints

Questions to Ask:

- When does the stakeholder need a deliverable? (specific date or relative)
- Hard deadline tied to a specific event, or flexible?
- Level of polish expected? (quick estimate vs. production-grade)
- One-time analysis or recurring/productionized?
- How much team time can be realistically allocated?

Timeline Benchmarks for Statistics Methods

Delivery Type	Typical Timeline	What's Realistic
Quick directional estimate	1-3 days	Descriptive stats, simple tests, existing data
Exploratory analysis / POC	1-3 weeks	Data cleaning, initial modeling, preliminary results
Validated statistical study	4-8 weeks	Full analysis, assumption checking, review cycles
Production recurring report	8-16 weeks	Automation, monitoring, documentation, handoff

Feasibility Signals — Timeline

- GREEN: Timeline aligns with delivery type; team has capacity
- YELLOW: Tight but achievable with scope reduction or dedicated resources
- RED: Incompatible with requested deliverable; needs negotiation

Dimension 3: Technical Constraints

- What compute environment is available? (local, cloud, shared cluster)
- Specific technology requirements or restrictions? (approved tools/languages)
- Does the solution need to integrate with existing systems?
- Latency requirements? (batch OK vs. near-real-time)
- GREEN: Existing infrastructure and tools support the approach
- YELLOW: Some setup needed; achievable within timeline
- RED: Required infrastructure doesn't exist or can't be provisioned

Dimension 4: Stakeholder & Output Expectations

This dimension is critical because how the stakeholder plans to use the output directly impacts method selection.

- Who is the end user? (analyst, executive, automated system, end customer)
- What format? (dashboard, report, API, email alert, presentation)
- How will the user interact? (view only, manipulate parameters, real-time)
- Will users need "what-if" scenarios?
- What level of accuracy or confidence is required?
- Does the stakeholder understand the limitations of statistical analysis here?

Why Output Format Matters for Method Selection

User Interaction	Implications for Method Choice
Static report / one-time presentation	Any method works; interpretability is key
Living dashboard with regular updates	Method must be automatable and reproducible
Parameter manipulation / what-if	Must support scenario modeling; parametric preferred
Real-time or near-real-time output	Must be computationally efficient; pre-computed preferred
Feeding into another system or model	Output format/schema must be defined; API needed

- GREEN: Output expectations clear and achievable
- YELLOW: Some expectation-setting needed
- RED: Expectations fundamentally misaligned with what analysis can deliver

Dimension 5: Organizational & Process Constraints

- Data privacy or compliance requirements? (PII, GDPR, internal policies)
- Does the approach require legal, security, or governance approval?
- Ethical considerations? (bias in data, fairness across groups)
- Has anyone else solved this or a similar problem?
- Critical institutional knowledge that's not documented?

- GREEN: No significant organizational blockers
- YELLOW: Approvals needed but achievable
- RED: Compliance/governance blockers that can't be resolved in time

Overall Feasibility Assessment

Rating	Meaning	Recommended Action
FEASIBLE	All GREEN or YELLOW with manageable risk	Proceed to method selection
CONDITIONALLY FEASIBLE	One or more YELLOW with significant risk	Proceed with scope reduction or timeline extension; document risk

NOT FEASIBLE AS DEFINED	One or more RED dimensions	Return to problem framing; negotiate scope, timeline, or data
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Template Summary Statement

"Based on our assessment, this use case is [RATING]."

- **Data Wish List vs. Reality:** Needed [list] / Available [list] / Gaps [list and impact]
- **Key Constraints:** [timeline, technical, organizational]
- **Recommended Mitigations:** [if applicable]
- **Recommendation:** [proceed / proceed with caveats / reframe]